Electricity Consumption Forecasting with Recurrent Architectures

LSTM Baseline - Progress Report

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1 GitHub Repository

The repository for this project is available at:

- `https://github.com/eerdemYlmzz/electricity-consumption-forecasting`

2 Dataset

2.1 Source

The dataset used in this study is the *Power Consumption of Tetouan City* dataset.

- Kaggle: `https://www.kaggle.com/datasets/fedesoriano/electric-power-consumption`
- UCI Machine Learning Repository: `https://archive.ics.uci.edu/dataset/849/power+consumption+of+tetouan+city`

2.2 Related Publication

A paper associated with this dataset was found:

> A. Salam and A. E. Hibaoui, "Comparison of Machine Learning Algorithms for the Power Consumption Prediction: — Case Study of Tetouan city —," *2018 6th International Renewable and Sustainable Energy Conference (IRSEC)*, Rabat, Morocco, 2018, pp. 1–5, doi: 10.1109/IRSEC.2018.8703007.

2.3 Original Features

The original dataset contains the following variables:

- Temperature
- Humidity
- Wind Speed
- General Diffuse Flows
- Diffuse Flows

- Power Consumption — Zone 1
- Power Consumption — Zone 2
- Power Consumption — Zone 3

Only the first power consumption feature (**Zone 1**) is used in this study.

2.4 Data Used in This Study

Property	**Value**
Shape	52416×3
`datetime`	year, month, day
`time`	10-minute stride, starting at 00:00; next day follows 23:50 + 10
`consumption`	consumption value, e.g. 34.0556962 kWh
Date range	2025-07-01 to 2026-06-29 (nearly one year)
Scaling / normalization	None applied

Table 1: Structure of the data used for training.

2.5 Date Modification

The dates in the current data differ from the original dataset. The original date range was **1/1/2017 to 31/12/2017**. The date component was dropped by the following preprocessing step, which keeps only the hour and minute:

```
df["time"] = df["time"].apply(lambda x: pd.to_datetime(x).strftime("%H:%M"))
```

3 Model Architecture and Training Setup

Parameter	**Value**
Sequence length	60
Input size	1
Hidden size	64
Stacked layers	2
Batch size	64
Shuffle	False
Loss function	MSELoss
Optimizer	Adam
Learning rate	1×10^{-3}
Epochs	10

Table 2: Model and training hyperparameters.

3.1 Train / Test Split

Split	**Shape** (samples, sequence length, features)	**Ratio**
Train	$41884 \times 60 \times 1$	0.8
Test	$10472 \times 60 \times 1$	0.2

Table 3: Train and test set sizes.

4 Training Results

Epoch	Train Loss
1	257.8208
2	49.5708
3	50.0217
4	29.0431
5	1.8884
6	0.6231
7	0.3642
8	0.2750
9	0.2347
10	0.2144

Table 4: Training loss per epoch (10 epochs).

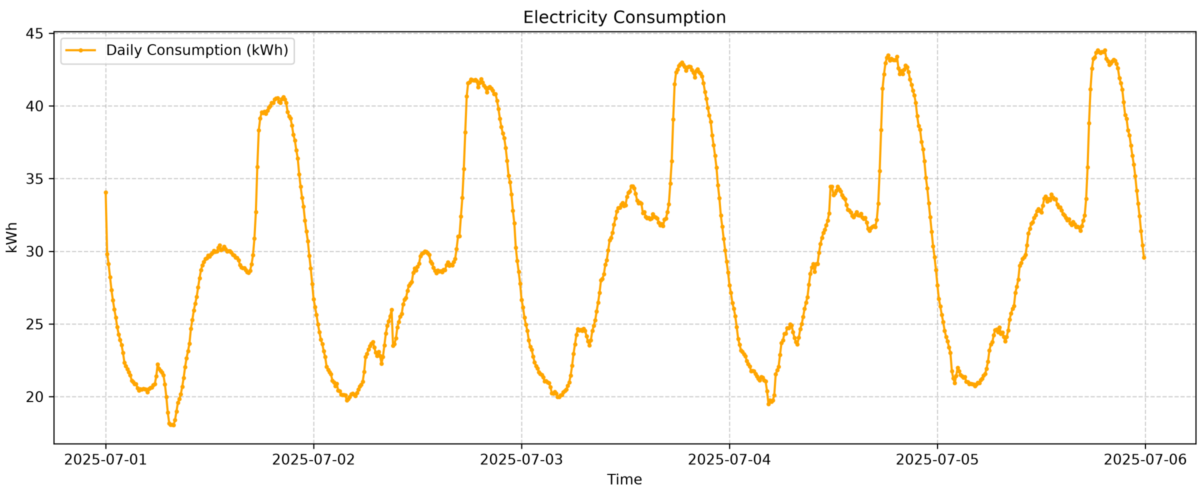

Figure 1: Electricity consumption.

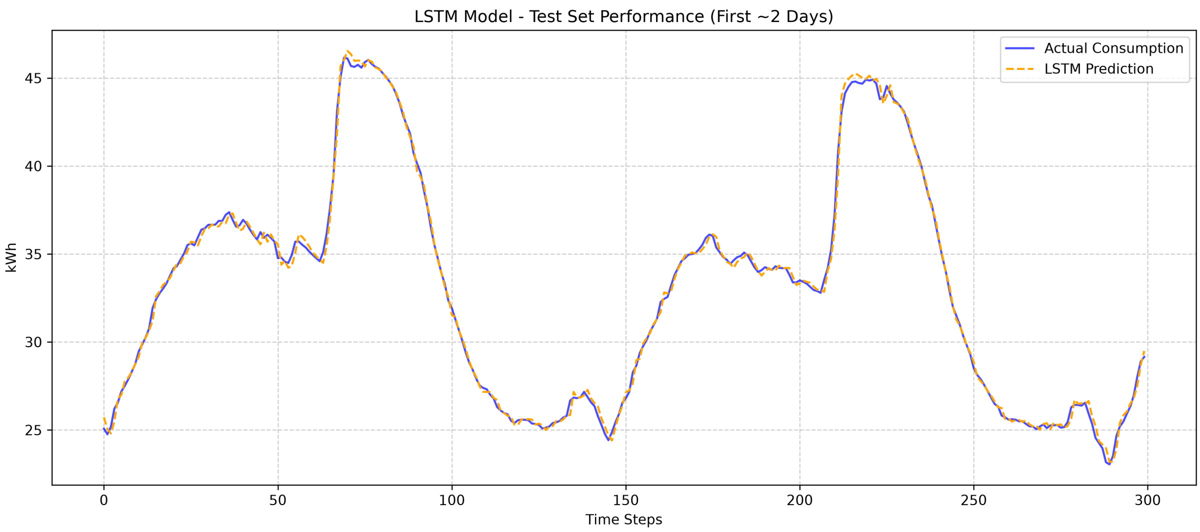

Figure 2: LSTM model — test set performance.

5 Outputs

- Figures saved at `dpi = 300`.
- Hyperparameters saved to `hyperparameter_logs.json`.
- Model saved to `lstm_model_1.pth`.

6 Planned Work

- GRU
- BiGRU
- Attention...
- TCN
- Newer attention architectures (Kimi K3, DeepSeek V3)
- Sequential data modeling

7 Open Questions and Missing Parts

- **Numeric precision.** Consumption values such as 34.0556962 — do all numbers hold 8 bytes of memory?
- **No validation set.** There is currently no validation split.
- **No early stopping.**
- **No validation loss per epoch.** Only training loss is recorded.
- **Attention and residual types.** Which attention / residual variants should be used (Kimi K3s, DeepSeek V3s)?
- **No scaling or normalization** is currently applied to the data.
- **Single feature only.** Only Zone 1 power consumption is used; temperature, humidity, wind speed, general diffuse flows, diffuse flows and Zones 2–3 are unused.